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Impact of maternal age on birth weight-related adverse outcomes in newborns: a retrospective study in south-central China

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Abstract

Background Birth weight is a critical indicator for assessing fetal development and newborn health status. This study aimed to examine both linear and nonlinear associations between maternal age and birth weight and their related adverse outcomes.

Methods 15,923 delivery data from 2018 to 2021 for pregnant women from the Changsha Maternal and Child Health Care Hospital were reviewed by a retrospective study. Basic information and infant birth weight were retrieved from the Medical Birth Registry. Multivariable regression models and the restricted cubic splines (RCS) analysis were used to identify the associations between maternal age and birth weight and its related adverse outcomes.

Results The ages of pregnant women trended upward from 2018 to 2021. Maternal age had a nonlinear association with birth weight (P for nonlinear = 0.028). Gestational weeks had a masking effect on the association between maternal age and birth weight, with an effect value of -7.368. A nonlinear association was found between maternal age and macrosomia (P for nonlinear = 0.009). Maternal age increased the risk for large for gestational age (LGA) (OR = 1.016, 95% CI: 1.006–1.027, P = 0.002) and preterm birth (OR = 1.028, 95% CI: 1.008–1.049, P = 0.005).

Conclusions This retrospective study from south-central China indicates that pregnant women are getting older and maternal age is nonlinearly associated with birth weight, while this association is masked by gestational weeks. Advanced maternal age increases the risk for heavier birth weight-related adverse outcomes.

Keywords Birthweight, Maternal age, Gestational weeks, Adverse outcomes

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Introduction

Birth weight is a critical indicator for assessing foetal development and newborn health status [1]. Abnormal birth weights are linked to various adverse outcomes including low birth weight, macrosomia, and being small (SGA) or large (LGA) for gestational age, as well as pre-term birth. These conditions not only impact children's immediate and long-term health, manifesting as stunted growth, mental retardation, and increased childhood mortality, but also increase the risk for chronic diseases such as hypertension and diabetes in adulthood [2–4].

Factors influencing birth weight include preexisting medical conditions, obstetrical history [5–7], and maternal social characteristics [8]. Numerous studies have identified maternal age as a key factor, particularly noting that advanced maternal age, typically > 35 years, is a risk factor for adverse birth outcomes such as premature birth and low birth weight [9–11]. In developed countries, the delay in childbearing has been significant, and in China, this trend has been influenced by the “second-child” policy initiated in 2016 [12]. Maternal age at childbirth may reflect various physiological and social processes, some of which might positively influence the prevention of adverse birth outcomes [13].

While many studies support the association between maternal age and birth weight, most have focused on the effects of extreme maternal ages on adverse outcomes [14, 15]. The precise nature of these associations, whether linear or nonlinear, remains unclear, complicating the development of effective intervention strategies. Furthermore, it is uncertain whether the relationship between maternal age and birth weight is directly due to age or a combination of associated factors. Recent research suggests that maternal age alone may not independently predict low birth weight, indicating that other, unobserved factors could confound this association [16].

In this study, we examined both linear and nonlinear associations between maternal age at childbirth and birth weight and its related adverse outcomes, and we identified confounding factors through a retrospective study in south-central China.

Methods

Study population

This retrospective study was conducted at the Changsha Maternal and Child Health Care Hospital, located in south-central China. It included all women who delivered at this hospital from January 2018 to December 2021. Inclusion criteria were pregnant women of gestational age ≥ 20 weeks and delivering live births at the hospital. The study excluded cases with congenital anomalies. In total, the records of 15,293 deliveries were reviewed from January 2018 to December 2021. The study was approved by the Ethics Committee of the Changsha City Maternal

and Child Health Care Hospital (No. EC-20230111-04). The data from a retrospective study are de-identified, and the requirement for informed consent was therefore waived by the Ethics Committee of the Changsha City Maternal and Child Health Care Hospital. All methods in this study complied with the Helsinki Declaration.

Birth weight-related adverse outcomes

The main adverse outcomes related to birth weight included low birth weight, macrosomia, SGA, LGA, and preterm birth. Information on infant birth weight and gestational weeks was retrieved from the Medical Birth Registry, developed by the local hospital and recorded by nurses or obstetricians. Low birth weight was defined as a birth weight < 2500 g, and macrosomia was defined as a birth weight ≥ 4000 g. SGA and LGA were assessed based on birth weight and gestational age standards for male and female newborns of various gestational ages across 15 cities in China [17]. Preterm birth was defined as a gestational age at birth < 37 weeks.

Associated variables

Basic information including maternal age, gestational weeks, residence (city or rural), occupation (farming or migrant workers, public officials, enterprise personnel, businessmen, unemployed, or other), education (junior and below, high/technical secondary school, or college and above), parity (primiparas or multiparas), infant birth weight, sex, and birth years (2018, 2019, 2020, or 2021), was reviewed from the Medical Birth Registry.

Statistical analyses

The data were conducted using SPSS software for further analysis (version 13.0, Chicago, IL, USA). Continuous variables are presented as mean $\pm$ standard deviation (SD), and categorical variables are reported as frequencies or percentages. For continuous variables and categorical variables the following statistical tests were employed: t-test and Chi-Square Tests. Multivariable linear regression models were used to identify the associations between maternal age and birth weight. Model 1 was established without any adjustments; Model 2 was adjusted for residence, occupation, education, infant sex, birth year, and parity. The standardised coefficients (β) and 95% confidence intervals (CI) were calculated to determine the strengths of the associations. Multivariable logistic regression models were used to examine the associations between maternal age and adverse birth outcomes. Model 1 was established without any adjustments; Model 2 was adjusted for residence, occupation, education, infant sex, birth year, and parity. The odds ratios (OR) and 95% CI were calculated to determine the strengths of the associations. $P < 0.05$ was considered statistically significant, using a two-sided test. The

mediation effect of related variables between maternal age and birth weight was assessed by the bootstrap test model, conducted using SPSSAU. A restricted cubic spline (RCS) analysis was used to evaluate the nonlinear relationship between maternal age and birth weight and its related adverse outcomes. RCS analyses were performed using R software, version 4.2.2, provided by the R Project for Statistical Computing.

Results

General characteristics of participants

Table 1 presents the general characteristics of the participants. In all, 15,923 pregnant women with a mean age of 31 years (SD=4 years) were included. The mean gestational weeks was 39 weeks (SD=1 week). The infants had an average birth weight of 3268 g (SD=456 g). The prevalence of low birth weight, macrosomia, SGA, LGA, and preterm birth were 4.03%, 5.19%, 2.76%, 24.91%, and 5.23%, respectively. There were significant differences in birth weight and gestational weeks among all adverse

birth outcomes. Significant differences among all adverse birth outcomes, except for macrosomia and SGA, were found for maternal age. Mothers with a higher educational level had lower incidences of low birth weight and preterm birth. Maternal multiparas had higher rates of low birth weight, preterm birth, and LGA. Males had higher rates of macrosomia and LGA but lower rates of low birth weight and SGA. In addition, the ages of pregnant women trended upward from 2018 to 2021. The most frequent maternal age increased from 28 years old in 2018 to 31 years old in 2021 (Fig. 1(A)).

Association between maternal age and birth weight

Figure 1(B) shows the results from the linear regression model assessing the association between maternal age and birth weight. No associations were observed between maternal age and birth weight in the unadjusted model or in the model adjusted for residence, occupation, education, infant sex, birth year, and parity.

Table 1 The general characteristics of the participants

Characteristics	Total (N)	Low birth weight (N(%))		Macrosomia (N(%))		SGA (N(%))		LGA (N(%))		Preterm birth (N(%))	
Birth weight (mean(sd), g)	3268 (456)	2082	383**	4154	196**	2432	371**	3772	270**	2364	538**
Maternal age (mean(sd), years)	31 (4)	31	(4)**	31	(4)	31	(4)	31	(4)**	32	(4)**
Gestational week (mean(sd), weeks)	39 (1)	35	(3)**	40	(1)**	38	(2)*	39	(1)**	35	(2)**
Residence											
City	11,532	427	(3.70)*	615	(5.33)	316	(2.74)	2885	(25.02)	560	(4.86)*
Rural	4391	214	(4.87)	211	(4.81)	124	(2.82)	1081	(24.62)	272	(6.19)
Occupation											
Farming	1181	64	(5.42)*	58	(4.91)	49	(4.15)	283	(23.96)*	75	(6.35)
Public officials	2820	99	(3.51)	138	(4.89)	75	(2.66)	665	(23.58)	130	(4.61)
Enterprises personnel	3155	107	(3.39)	162	(5.13)	78	(2.47)	730	(23.14)	157	(4.98)
Businessmen	1043	42	(4.03)	57	(5.47)	29	(2.78)	306	(29.34)	57	(5.47)
Unemployment	2168	92	(4.24)	109	(5.03)	55	(2.54)	561	(25.88)	119	(5.49)
Other	5459	231	(4.23)	296	(5.42)	151	(2.77)	1394	(25.54)	288	(5.28)
Education											
College and above	12,360	462	(3.74)*	650	(5.26)	340	(2.75)	3033	(24.54)	615	(4.98)*
High or technical secondary school	2494	126	(5.05)	129	(5.17)	72	(2.89)	645	(25.86)	150	(6.01)
Junior and below	885	45	(5.08)	39	(4.41)	24	(2.71)	236	(26.67)	57	(6.44)
Parity											
Primiparas	8296	227	(2.74)**	420	(5.06)	270	(3.25)**	1735	(20.91)**	300	(3.62)**
Multiparas	7627	414	(5.43)	406	(5.32)	170	(2.23)	2231	(29.25)	532	(6.98)
Infant gender											
Boy	8290	296	(3.57)*	551	(6.65)**	254	(3.06)*	2291	(27.64)**	452	(5.45)
Girl	7633	345	(4.52)	275	(3.60)	186	(2.44)	1675	(21.94)	380	(4.98)
Birth year											
2018	3701	147	(3.97)	202	(5.46)*	107	(2.89)	973	(26.29)*	188	(5.08)
2019	4184	151	(3.61)	239	(5.71)	118	(2.82)	1055	(25.22)	201	(4.80)
2020	4014	156	(3.89)	208	(5.18)	99	(2.47)	1028	(25.61)	200	(4.98)
2021	4024	187	(4.65)	177	(4.40)	116	(2.88)	910	(22.61)	243	(6.04)
Total	15,923	641	(4.03)	826	(5.19)	440	(2.76)	3966	(24.91)	832	(5.23)

* $P < 0.05$; ** $P < 0.001$; comparing by T Test or Chi-Square Tests

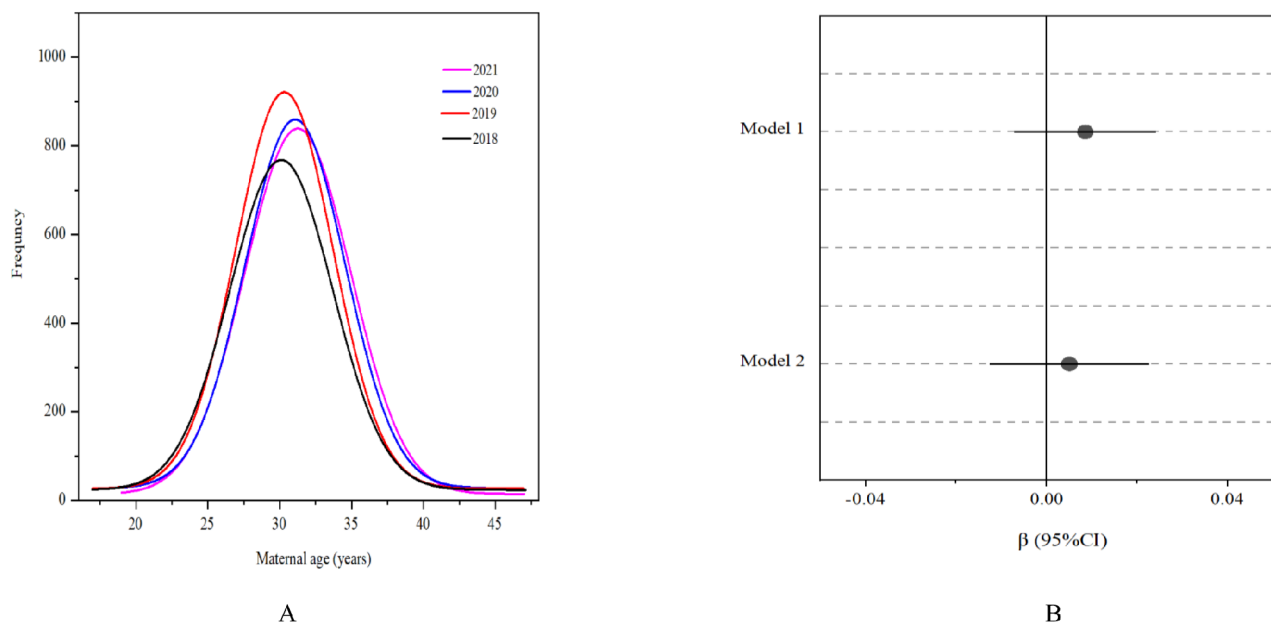


Fig. 1 (A) Frequency of maternal age from 2018 to 2021; (B) The association of maternal age and birth weight by multiple linear regression analysis. Model 1 was established without any adjustments; Model 2 was adjusted for residence, occupation, education, infant sex, birth year, and parity

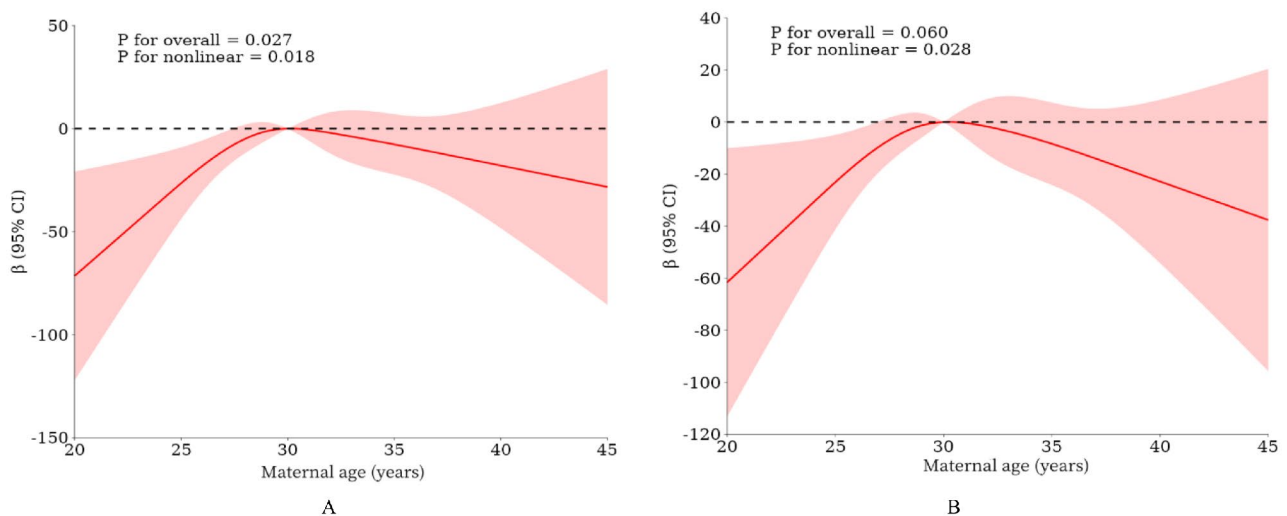


Fig. 2 The nonlinear relationships between maternal age and birth weight by restricted cubic spline approach. (A) Model 1 was established without any adjustments; (B) Model 2 was adjusted for residence, occupation, education, infant sex, birth year, and parity

Figure 2 shows the nonlinear association between maternal age and birth weight using RCS. A nonlinear association was observed in the unadjusted model (P for nonlinear = 0.018) and the model adjusted for residence, occupation, education, infant sex, birth year, and parity (P for nonlinear = 0.028).

Mediating effect of gestational weeks

Table 2 shows the results of analyses of the mediating effect of gestational weeks on the association between maternal age and birth weight. No significant differences

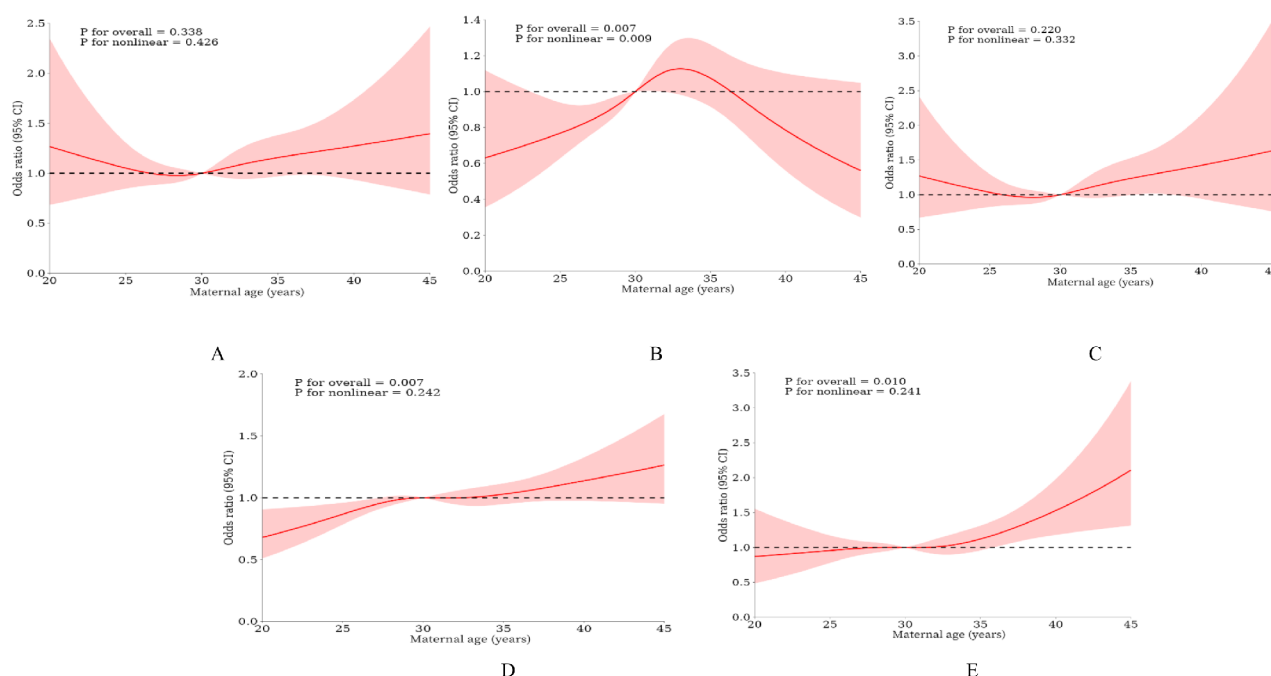
were found in the overall effect of maternal age on birth weight, with a direct effect value of 8.346 and a 95% CI of 6.901 to 9.790. In addition, gestational weeks had a masking effect on the association between maternal age and birth weight, with an effect value of -7.368 and a 95% CI for bootstrap of -0.073 to -0.050.

Association between maternal age and birth weight-related adverse outcomes

The results of RCS analyses indicated a nonlinear association between maternal age and macrosomia in the

Table 2 The mediating effect analysis of gestational week on the association between maternal age and birth weight

Group	Effect value	95% CI		z/t	p
		Lower	Upper		
Maternal age => Gestational weeks => Birth weight (Indirect effect)	-7.368	-0.073	-0.050	-1495.391	< 0.001
Maternal age => Gestational weeks	-0.041	-0.047	-0.036	-13.931	< 0.001
Gestational weeks => Birth weight	177.986	174.156	181.815	91.099	< 0.001
Maternal age => Birth weight (Direct effect)	8.346	6.901	9.790	11.323	< 0.001
Maternal age => Birth weight (Total effect)	0.977	-0.794	2.748	1.081	0.280

**Fig. 3** The nonlinear relationships between maternal age with adverse pregnancy outcome by restricted cubic spline approach. (A) Low birth weight; (B) Macrosomia; (C) SGA; (D) LGA; (E) Perterm birth. Adjusting for residence, occupation, education, infant gender, birth year, and parity

unadjusted model (P for nonlinear = 0.012, Supplemental 1) and model for adjusting for residence, occupation, education, infant sex, birth year, parity, and gestational week (P for nonlinear = 0.009, Fig. 3(B)). Moreover, maternal age was positively associated with low birth weight in the unadjusted model (Supplemental 1), but no association was found after adjusting for residence, occupation, education, infant sex, birth year, and parity (Fig. 3(A)). Positive associations were found between maternal age, and LGA and preterm birth in the unadjusted model and the model after adjusting for residence, occupation, education, infant sex, birth year, and parity (Supplemental 1 (D) and (E) and Fig. 3(D) and (E)).

Table 3 details the association between maternal age and adverse birth outcomes in the logistic regression model. Maternal age was a risk factor for low birth weight (OR = 1.047, 95% CI: 1.027–1.067, P < 0.001) in the unadjusted model, but no association was found after

adjusting for residence, occupation, education, infant sex, birth year, and parity. No association was found between maternal age and macrosomia. In addition, maternal age was a risk factor for LGA (OR = 1.016, 95% CI: 1.006–1.027, P = 0.002) and preterm birth (OR = 1.028, 95% CI: 1.008–1.049, P = 0.005) after adjusting for confounding factors.

Discussion

From this retrospective study of pregnant women in south-central China, we found the ages of pregnant women trended upward from 2018 to 2021. Maternal age had a nonlinear association with birth weight, and this association was masked by gestational weeks. Moreover, we also confirmed that maternal age had a nonlinear association with macrosomia, and increased the risk of LGA, and preterm birth.

Table 3 The association of maternal age with adverse pregnancy outcome by logistic regression analysis

Adverse outcomes	OR (95%CI)	P
Low birth weight		
Model 1	1.047(1.027,1.067)	< 0.001
Model 2	1.015(0.993,1.037)	0.195
Macrosomia		
Model 1	1.016(0.998,1.034)	0.077
Model 2	1.015(0.995,1.036)	0.132
SGA		
Model 1	0.995(0.971,1.019)	0.660
Model 2	1.021(0.993,1.049)	0.138
LGA		
Model 1	1.037(1.028,1.047)	< 0.001
Model 2	1.016(1.006,1.027)	0.002
Perterm birth		
Model 1	1.057(1.039,1.075)	< 0.001
Model 2	1.028(1.008,1.049)	0.005

Model 1 was established without any adjustments; Model 2 was adjusted for residence, occupation, education, infant sex, birth year, and parity

Our study used the RCS analysis and found a nonlinear relationship between maternal age and birth weight. This supports many previous studies that have reported that maternal age increases both the risk for low birth weight and macrosomia [18, 19]. However, we also found no association between maternal age and birth weight in the linear regression model. The RCS analysis can provide the dose-response relations of related variables [20]. This result provided a new insight to precisely prevent birth weight-related adverse outcomes.

An association between birth weight-related adverse outcomes and advanced maternal age has been widely reported [16, 21, 22]. However, some studies have reported a U-shaped relationship; thus, both atypically young and atypically old maternal ages may increase the risk for low birth weight [23, 24]. In our study, advanced maternal age increased the risk of low birth weight in the crude model, similar to the results of most studies. However, this association disappeared for controlling demographic factors, consistent with Goisis et al. [16]. In addition, we observed no significant association between maternal age and SGA, consistent with Zhou et al. [5]. They considered several confounding factors such as nutritional or socioeconomic factors may contribute to the risk of SGA. These results suggest that there are several confounding factors that are related to both maternal age and birth weight-related adverse outcomes.

Furthermore, we found that older maternal age increased the risk for LGA. Birth weight is mainly based on pre-pregnancy weight and weight gain during pregnancy. Older women may have higher pre-pregnancy BMI and excessive gestational weight gain, which may lead to heavier birth weight and increase the risk for its adverse outcomes, such as LGA or macrosomia [5, 25].

Moreover, older pregnant women suffer from a high prevalence of abnormal glucose metabolism and gestational diabetes due to excessive nutritional intake, which may result in an increase in the possibility of LGA or macrosomia [18].

We also found that maternal age was positively associated with preterm birth, consistent with most prior studies [16, 26]. Such studies have suggested that advanced maternal age increases the likelihood of preterm birth due to a lower quality of oocytes, weakening of the placenta, and a heightened risk of pregnancy complications [27, 28]. The masking effect of gestational weeks in this study may be explained by this finding, as fetuses have more time to grow and develop, leading to heavier newborns [25]. However, while maternal age increased the risk for preterm birth, it showed a negative association with gestational age, which could mask the positive impact of maternal age on birth weight. The nonlinear association of maternal age with macrosomia also can be explained by this effect. This suggests that newborns face a higher risk for adverse outcomes related to heavier birth weight, aside from the masking effect of gestational weeks.

The major strengths of this study were that we focused on both linear and nonlinear associations between maternal age and the risk of adverse birth outcomes. Maternal age was nonlinearly associated with birth weight and increased the risk for LGA and preterm birth. This information provides detailed information that can be used to prevent birth weight-related adverse outcomes.

Our study also had several limitations. First, the records of the basic information (e.g. mother's occupation, education level, and parity) may have biased effect estimates due to the retrospective design. Second, our result found gestational weeks masked the association of maternal age with birth weight. Nevertheless, gestational weeks are affected by medical care, especially for delivery methods. A definite proportion of women were not delivered naturally, associated with maternal or fetal distress or concern about an extended gestation placing the fetus at risk of stillbirth, etc [29, 30]. These factors may limit the application of our results. The masked effect in women with natural childbirth may be an interesting theme in future studies. Finally, although detailed information about many potential sociodemographic factors associated with mothers and infants had been considered. Maternal pregnancy weight associated with both maternal age and birth weight [5, 31], were not included. The residual confounding from this factor needs to be further explored in future research.

Conclusion

This retrospective study from south-central China indicates that pregnant women are getting older and maternal age is nonlinearly associated with birth weight, while this association is masked by gestational weeks. Maternal age had a nonlinear association with macrosomia and increased the risk of LGA, and preterm birth. The higher risk of heavier birth weight-related adverse outcomes for advanced pregnant women should be paid more attention by obstetricians.

Abbreviations

SGA	Small for gestational age
LGA	Large for gestational age
SD	Standard deviation
RCS	Restricted cubic splines
β	Standardised coefficients
CI	Confidence interval
OR	Odds ratio

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12884-025-07199-7>.

Supplementary Material 1

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Not applicable.

Author contributions

Z.H. and A.Z. performed the literature search, analyzed and interpreted the data, and drafted the manuscript. Y.Z., J.W. and X.T. collected, verified, and interpreted the data. A.Z. contributed to the design of this study and data interpretation. All authors reviewed and approved the final version of the manuscript.

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Data availability

The datasets generated and/or analysed during the current study are not publicly available due information privacy but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of the Changsha City Maternal and Child Health Care Hospital (No. EC-20230111-04). The data from a retrospective study are de-identified, and the requirement for informed consent was therefore waived by the Ethics Committee of the Changsha City Maternal and Child Health Care Hospital. All methods in this study complied with the Helsinki Declaration.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Bonanni G, Airolidi C, Berghella V. Birthweights at term have increased globally: insights from a systematic review of 183 million births. *Am J Obstet Gynecol* 2024.
2. Blencowe H, Lee ACC, Cousens S, Bahalim A, Narwal R, Zhong N, Chou D, Say L, Modi N, Katz J, et al. Preterm birth-associated neurodevelopmental impairment estimates at regional and global levels for 2010. *Pediatr Res*. 2013;74(S1):17–34.
3. Rashti R, Ghasemi F, Poorolajal J. Association between birth weight and risk of nonneurological childhood cancers: a systematic review and meta-analysis. *Eur J Cancer Prev* 2024.
4. Mohseni R, Mohammed SH, Safabakhsh M, Mohseni F, Monfared ZS, Seyyedi J, Mejahreh ZN, Alizadeh S. Birth Weight and Risk of Cardiovascular Disease incidence in Adulthood: a dose-response Meta-analysis. *Curr Atheroscler Rep*. 2020;22(3):12.
5. Zhou Y, Yin S, Sheng Q, Yang J, Liu J, Li H, Yuan P, Zhao Y. Association of maternal age with adverse pregnancy outcomes: a prospective multicenter cohort study in China. *J Global Health*. 2023;13:04161.
6. Zhang Y, Jin Y, Duan W, Yu H, Yuan H. Trend and influencing factors of low birth weight in Jiading District of Shanghai from 2011 to 2021. *J Prev Med Inform*. 2023;39(6):693–703.
7. Hinkle SN, Albert PS, Mendola P, Sjaarda LA, Yeung E, Boghossian NS, Laughon SK. The Association between Parity and Birthweight in a longitudinal consecutive pregnancy cohort. *Paediatr Perinat Epidemiol*. 2013;28(2):106–15.
8. Hidalgo-Lopezosa P, Jiménez-Ruz A, Carmona-Torres JM, Hidalgo-Maestre M, Rodríguez-Borrego MA, López-Soto PJ. Sociodemographic factors associated with preterm birth and low birth weight: a cross-sectional study. *Women Birth*. 2019;32(6):e538–43.
9. Esposito G, Mauri PA, Cipriani S, Franchi M, Corrao G, Parazzini F. The role of maternal age on the risk of preterm birth among singletons and multiples: a retrospective cohort study in Lombardy, Northern Italy. *BMC Pregnancy Childbirth*. 2022;22(1):234–45.
10. Cnattingius S, Forman MR, Berendes HW. Delayed childbearing and risk of adverse perinatal outcome: a population-based study. *JAMA*. 1992;268:886.
11. Luke B, Brown MB. Elevated risks of pregnancy complications and adverse outcomes with increasing maternal age. *Hum Reprod*. 2007;22(5):1264–72.
12. Tian M-L, Ma G-J, Du L-Y, Jin Y, Zhang C, Xiao Y-G, Tang Z-j. The Effect of 2016 Chinese second-child policy and different maternal age on pregnancy outcomes in Hebei Province, China. *BMC Pregnancy Childbirth*. 2023;23(1):267–76.
13. Stein Z. The risks of having children in later life. *BMJ*. 2000;320:1681–2.
14. Londero AP, Rossetti E, Pittini C, Cagnacci A, Driul L. Maternal age and the risk of adverse pregnancy outcomes: a retrospective cohort study. *BMC Pregnancy Childbirth*. 2019;19(1):261.
15. Lu B, Han B, Hu H, Long W, Wang L, Cai Z, Wang H, Yu B. The association of pregnant women's age and maternal and infant complications under the new fertility policy. *Chin J Perinat Med*. 2019;22(3):7.
16. Goisis A, Remes H, Barclay K, Martikainen P, Myrskylä M. Advanced maternal age and the risk of low Birth Weight and Preterm Delivery: a within-family analysis using Finnish Population registers. *Am J Epidemiol*. 2017;186(11):1219–26.
17. Scientific Collaboration Group on Physical Development of Newborns in 15 Cities of China. The correction report of birth weight on newborns with different gestational ages in 15 cities of China. *Chin J Practical Pediatr*. 1992;7(6):306.
18. Teng X-h, Pan S-l. The relationship among maternal age, risk factors on pregnancy outcomes: a retrospective cohort study. *J Practical Obstet Gynecol*. 2017;33(9):692–6.
19. Liu X, Zhang W. Effect of maternal age on pregnancy: a retrospective cohort study. *Chin Med J*. 2014;127(12):2241–6.
20. Gilboa SM, Correa A, Alverson CJ. Use of Spline Regression in an analysis of maternal prepregnancy body Mass Index and adverse birth outcomes: does it tell us more than we already know? *Ann Epidemiol*. 2008;18(3):196–205.
21. Aradhya S, Tegunimataka A, Kravdal Ø, Martikainen P, Myrskylä M, Barclay K, Goisis A. Maternal age and the risk of low birthweight and pre-term delivery: a pan-nordic comparison. *Int J Epidemiol*. 2023;52(1):156–64.
22. Ratiu D, Sauter F, Gilman E, Ludwig S, Ratiu J, Mallmann-Gottschalk N, Mallmann P, Gruttner B, Baek S. Impact of advanced maternal age on maternal and neonatal outcomes. *Vivo*. 2023;37(4):1694–702.
23. Yang M, Wang J. Retrospective cohort study of the effect of maternal age on adverse neonatal outcomes. *Chin J Evid Based Pediatr*. 2019;14(4):276–81.

24. Ning J, Chen L, Chen L, Huangpu X, Yu H. Relationship between maternal age and adverse birth outcomes and inspiration for appropriate age of childbearing. *Mod Prev Med*. 2021;48(20):3720–4.
25. Wang Y, Gong Y, Xu Y, Wang X, Shan S, Cheng G, Zhang B. Maternal age-specific risks for adverse birth weights according to gestational weight gain: a prospective cohort in Chinese women older than 30. *BMC Pregnancy Childbirth*. 2024;24(1):36.
26. Zhu Y, Yu H, Wang C, Wang W, Fu C. Risk prediction of low birth weight infants in Shanghai. *Shanghai J Prev Med*. 2023;35(6):564–72.
27. He X, Huang J, Li B, Wang X. Relationship between maternal age and gestational age & birth weight of offspring in 5936 frozen-thawed embryo transfer cycles. *J Reprod Med*. 2020;29(4):435–42.
28. Zhang J, Savitz DA. Maternal age and placenta previa: a population-based, case-control study. *Am J Obstet Gynecol*. 1993;168(2):641–5.
29. Masters RK, Tilstra AM, Coleman-Minahan K. Increases in Obstetric interventions and changes in Gestational Age distributions of U.S. births. *J Womens Health (Larchmt)*. 2023;32(6):641–51.
30. Wen T, Turitz AL. Optimizing Term Delivery and Mode of Delivery. *Clin Perinatol*. 2020;47(4):799–815.
31. Goldstein RF, Abell SK, Ranasinha S, Misso ML, Boyle JA, Harrison CL, Black MH, Li N, Hu G, Corrado F, et al. Gestational weight gain across continents and ethnicity: systematic review and meta-analysis of maternal and infant outcomes in more than one million women. *BMC Med*. 2018;16(1):153.

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